

Question ID: 4fe4fd7c

| Assessment | Test | Domain  | Skill            | Difficulty                                          |
|------------|------|---------|------------------|-----------------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Medium <div><div></div><div></div><div></div></div> |

Question

$c(x) = mx + 500$

A company’s total cost  $c(x)$ , in dollars, to produce x shirts is given by the function above, where m is a constant and  $x > 0$ . The total cost to produce 100 shirts is \$800. What is the total cost, in dollars, to produce 1000 shirts? (Disregard the \$ sign when gridding your answer.)

Rationale

The correct answer is 3500. The given information includes a cost, \$800, to produce 100 shirts. Substituting  $c(x) = 800$  and  $x = 100$  into the given equation yields  $800 = m \cdot 100 + 500$ . Subtracting 500 from both sides of the equation yields  $300 = m \cdot 100$ . Dividing both sides of this equation by 100 yields  $3 = m$ . Substituting the value of m into the given equation yields  $c(x) = 3x + 500$ . Substituting 1000 for x in this equation and solving for  $c(x)$  gives the cost of 1000 shirts:  $3(1000) + 500$ , or 3500.